

MAY 2024



FINAL PRESENTATION

How to check constraints without relying on AI

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Introduction

We are the C-C-Code-Factory, and in this class, we collected 50 states economy data (excluding Washington D.C.) to see the correlation between certain aspects such as inflation or crime rate with economic growth.

Background

Project 4, a two-week-project requires us to create PK/FK constraints in BigQuery and logical checks for validation.

We only removed the duplication with AI, resulting an inferior submission.

LLM for crime_data

[+ Code](#)
[+ Text](#)

```
import itertools, json, pandas, pandas_gbq
from google.cloud import bigquery
import vertexai
from vertexai.generative_models import GenerativeModel, Part
```

prompt = """Here is a list of crime data containing year and state name.

I want you to check if the state name is from one of 50 U.S. states. If it does, make the characters in the name all small case and remove any spaces or underscores or dashes. If I also want you to convert the year to an integer.

And then combine the converted year and the converted state name into a single string.

Finally, add a new columns called "primary_key" at the beginning of the table and store the newly formed combination into the corresponding cell.

Return the results as a properly formatted json object with only one json object per line.

Don't return the records which are not from one of 50 U.S. states.

Don't return any empty json objects.

Don't return an explanation for your answer.

I will give you a list with the following columns: year,population,state,crime,crime_against_persons,crime_against_property,crime_against_society,_data_source,_load_time

You you return a list with the following columns: primary_key,year,population,state,crime,crime_against_persons,crime_against_property,crime_against_society,_data_source,_load_time

Here are some sample runs:

I pass you:

"2020.0,1351313.0,Arizona,83933.0,18434.0,47279.0,18220.0,CrimeData,2025-02-07 03:07:32.577832 UTC"

"2020.0,668166.0,New_York,35006.0,8211.0,24482.0,2313.0,CrimeData,2025-02-07 03:07:32.577832 UTC"

"2022.0,671803.0,District_of_Columbia,72403.0,21196.0,45536.0,5671.0,CrimeData,2025-02-07 03:07:32.577832 UT"

You return:

{2020arizona,2020,1351313.0,arizona,83933.0,18434.0,47279.0,18220.0,CrimeData,2025-02-07 03:07:32.577832 UTC}

{"2020newyork,2020,668166.0,newyork,35006.0,8211.0,24482.0,2313.0,CrimeData,2025-02-07 03:07:32.577832 UTC"}

"""

sql = """select * from finance_economics_events_int.crime_data

where population is not null

"""

6 batches will be sent to LLM

states are: ['2020.0,1351313.0,Arizona,83933.0,18434.0,18434.0,18220.0,CrimeData,2025-02-07 03:07:32.577832+00:00\n 2020.0,1844146.0,Arkansas,232969.0,63287.0,63287.0,35259.0,Cr:0: batch starting with state 2020.0

resp_text:

```
{
  "primary_key": "2020arizona",
  "year": 2020,
  "population": 1351313.0,
  "state": "arizona",
  "crime": 83933.0,
  "crime_against_persons": 18434.0,
  "crime_against_property": 18434.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020arkansas",
  "year": 2020,
  "population": 1844146.0,
  "state": "arkansas",
  "crime": 232969.0,
  "crime_against_persons": 63287.0,
  "crime_against_property": 63287.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020colorado",
  "year": 2020,
  "population": 4044858.0,
  "state": "colorado",
  "crime": 327846.0,
  "crime_against_persons": 60379.0,
  "crime_against_property": 60379.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020connecticut",
  "year": 2020,
  "population": 2862137.0,
  "state": "connecticut",
  "crime": 111569.0,
  "crime_against_persons": 27784.0,
  "crime_against_property": 27784.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020delaware",
  "year": 2020,
  "population": 298909.0,
  "state": "delaware",
  "crime": 66230.0,
  "crime_against_persons": 16131.0,
  "crime_against_property": 16131.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020georgia",
  "year": 2020,
  "population": 2546081.0,
  "state": "georgia",
  "crime": 284980.0,
  "crime_against_persons": 80969.0,
  "crime_against_property": 80969.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020idaho",
  "year": 2020,
  "population": 966222.0,
  "state": "idaho",
  "crime": 70989.0,
  "crime_against_persons": 16515.0,
  "crime_against_property": 16515.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020illinois",
  "year": 2020,
  "population": 144795.0,
  "state": "illinois",
  "crime": 15001.0,
  "crime_against_persons": 5430.0,
  "crime_against_property": 5430.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020indiana",
  "year": 2020,
  "population": 1942576.0,
  "state": "indiana",
  "crime": 157157.0,
  "crime_against_persons": 42922.0,
  "crime_against_property": 42922.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020iowa",
  "year": 2020,
  "population": 2035794.0,
  "state": "iowa",
  "crime": 130550.0,
  "crime_against_persons": 30180.0,
  "crime_against_property": 30180.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020kansas",
  "year": 2020,
  "population": 1247919.0,
  "state": "kansas",
  "crime": 85470.0,
  "crime_against_persons": 23761.0,
  "crime_against_property": 23761.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020kentucky",
  "year": 2020,
  "population": 1831363.0,
  "state": "kentucky",
  "crime": 158548.0,
  "crime_against_persons": 34343.0,
  "crime_against_property": 34343.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020louisiana",
  "year": 2020,
  "population": 156619.0,
  "state": "louisiana",
  "crime": 65819.0,
  "crime_against_persons": 17980.0,
  "crime_against_property": 17980.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020maine",
  "year": 2020,
  "population": 466254.0,
  "state": "maine",
  "crime": 27770.0,
  "crime_against_persons": 8205.0,
  "crime_against_property": 8205.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020maryland",
  "year": 2020,
  "population": 59008.0,
  "state": "maryland",
  "crime": 30334.0,
  "crime_against_persons": 5125.0,
  "crime_against_property": 5125.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020massachusetts",
  "year": 2020,
  "population": 6993995.0,
  "state": "massachusetts",
  "crime": 244872.0,
  "crime_against_persons": 68369.0,
  "crime_against_property": 68369.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020michigan",
  "year": 2020,
  "population": 6937365.0,
  "state": "michigan",
  "crime": 423874.0,
  "crime_against_persons": 152516.0,
  "crime_against_property": 152516.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020minnesota",
  "year": 2020,
  "population": 2007594.0,
  "state": "minnesota",
  "crime": 130373.0,
  "crime_against_persons": 24900.0,
  "crime_against_property": 24900.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020mississippi",
  "year": 2020,
  "population": 575244.0,
  "state": "mississippi",
  "crime": 54070.0,
  "crime_against_persons": 13720.0,
  "crime_against_property": 13720.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020missouri",
  "year": 2020,
  "population": 1948766.0,
  "state": "missouri",
  "crime": 211811.0,
  "crime_against_persons": 49525.0,
  "crime_against_property": 49525.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020montana",
  "year": 2020,
  "population": 518813.0,
  "state": "montana",
  "crime": 53990.0,
  "crime_against_persons": 13548.0,
  "crime_against_property": 13548.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020nebraska",
  "year": 2020,
  "population": 345759.0,
  "state": "nebraska",
  "crime": 27204.0,
  "crime_against_persons": 6551.0,
  "crime_against_property": 6551.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020nevada",
  "year": 2020,
  "population": 410368.0,
  "state": "nevada",
  "crime": 32194.0,
  "crime_against_persons": 8663.0,
  "crime_against_property": 8663.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020newhampshire",
  "year": 2020,
  "population": 1299709.0,
  "state": "newhampshire",
  "crime": 49273.0,
  "crime_against_persons": 14789.0,
  "crime_against_property": 14789.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020newmexico",
  "year": 2020,
  "population": 676407.0,
  "state": "newmexico",
  "crime": 80548.0,
  "crime_against_persons": 20426.0,
  "crime_against_property": 20426.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020newyork",
  "year": 2020,
  "population": 668166.0,
  "state": "newyork",
  "crime": 35006.0,
  "crime_against_persons": 8211.0,
  "crime_against_property": 8211.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020northcarolina",
  "year": 2020,
  "population": 5226511.0,
  "state": "northcarolina",
  "crime": 531141.0,
  "crime_against_persons": 121553.0,
  "crime_against_property": 121553.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020northdakota",
  "year": 2020,
  "population": 55616.0,
  "state": "northdakota",
  "crime": 49231.0,
  "crime_against_persons": 10745.0,
  "crime_against_property": 10745.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020ohio",
  "year": 2020,
  "population": 6267855.0,
  "state": "ohio",
  "crime": 334496.0,
  "crime_against_persons": 143338.0,
  "crime_against_property": 143338.0,
  "crime_against_society": 35259.0,
  "primary_key": "2020oklahoma",
  "year": 2020,
  "population": 1938408.0,
  "state": "oklahoma",
  "crime": 154675.0,
  "crime_against_persons": 35821.0,
  "crime_against_property": 35821.0,
  "crime_against_society": 35259.0
}
```

ison obis: [{ 'primary_key': '2020arizona', 'year': 2020, 'population': 1351313.0, 'state': 'arizona', 'crime': 83933.0, 'crime_against_persons': 18434.0, 'crime_against_property': 18434.0, 'crime_against_society': 35259.0 }

Background

As you can see in these screenshots, we used LLM to generate unique PK for each table. The generating time for each table is around 15 minutes, which is too long.

And we lose points for not doing PK/FK correctly.

Challenge

Project 6 requires us to transport these PK/FKs into dbt.

- For the primary key, foreign key, uniqueness, not null and relationship tests, I would suggest copying one of the `schema.yml` files from the snippets repo (such as [this one](#)) and using it as your starting point.

Challenges

- We need to makeup something we didn't do properly
- Using LLM in dbt is very difficult
- Final season – time management

Thoughts

50

Texas – texas

New York – New_York

NewYork

newyork

Row	year	state	population	crime	crime_against_p	crime_against_p	crime_against_s	_data_source	_load_time
1	2020	Pennsylvania	12504.0	156.0	22.0	98.0	36.0	CrimeData	2025-02-07 03:06:20.701827 U...
2	2020	Pennsylvania	12504.0	156.0	22.0	98.0	36.0	CrimeData	2025-02-07 03:07:32.577832 U...
3	2020	Wyoming	35257.0	2110.0	693.0	973.0	444.0	CrimeData	2025-02-07 03:07:32.577832 U...
4	2020	Wyoming	35257.0	2110.0	693.0	973.0	444.0	CrimeData	2025-02-07 03:06:20.701827 U...
5	2020	District_of_Columbia	0.0	2535.0	1062.0	1065.0	408.0	CrimeData	2025-02-07 03:07:32.577832 U...
6	2020	District_of_Columbia	0.0	2535.0	1062.0	1065.0	408.0	CrimeData	2025-02-07 03:06:20.701827 U...
7	2020	Vermont	362205.0	17845.0	3980.0	12470.0	1395.0	CrimeData	2025-02-07 03:06:20.701827 U...
8	2020	Vermont	362205.0	17845.0	3980.0	12470.0	1395.0	CrimeData	2025-02-07 03:07:32.577832 U...
9	2020	Maryland	59008.0	30334.0	5125.0	22846.0	2363.0	CrimeData	2025-02-07 03:07:32.577832 U...
10	2020	Maryland	59008.0	30334.0	5125.0	22846.0	2363.0	CrimeData	2025-02-07 03:06:20.701827 U...
11	2020	Illinois	144795.0	15001.0	5430.0	8537.0	1034.0	CrimeData	2025-02-07 03:06:20.701827 U...
12	2020	Illinois	144795.0	15001.0	5430.0	8537.0	1034.0	CrimeData	2025-02-07 03:07:32.577832 U...
13	2020	Nebraska	345759.0	27204.0	6551.0	14613.0	6040.0	CrimeData	2025-02-07 03:06:20.701827 U...
14	2020	Nebraska	345759.0	27204.0	6551.0	14613.0	6040.0	CrimeData	2025-02-07 03:07:32.577832 U...
15	2020	Maine	466254.0	27770.0	8205.0	16445.0	3120.0	CrimeData	2025-02-07 03:07:32.577832 U...
16	2020	Maine	466254.0	27770.0	8205.0	16445.0	3120.0	CrimeData	2025-02-07 03:06:20.701827 U...
17	2020	New_York	668166.0	35006.0	8211.0	24482.0	2313.0	CrimeData	2025-02-07 03:06:20.701827 U...
18	2020	New_York	668166.0	35006.0	8211.0	24482.0	2313.0	CrimeData	2025-02-07 03:07:32.577832 U...
19	2020	Nevada	410368.0	32194.0	8663.0	19474.0	4057.0	CrimeData	2025-02-07 03:07:32.577832 U...
20	2020	Nevada	410368.0	32194.0	8663.0	19474.0	4057.0	CrimeData	2025-02-07 03:06:20.701827 U...
21	2020	Rhode_Island	1065374.0	37731.0	9114.0	25125.0	3492.0	CrimeData	2025-02-07 03:07:32.577832 U...
22	2020	Rhode_Island	1065374.0	37731.0	9114.0	25125.0	3492.0	CrimeData	2025-02-07 03:06:20.701827 U...
23	2020	North_Dakota	555616.0	49231.0	10745.0	25514.0	12972.0	CrimeData	2025-02-07 03:07:32.577832 U...
24	2020	North_Dakota	555616.0	49231.0	10745.0	25514.0	12972.0	CrimeData	2025-02-07 03:06:20.701827 U...
25	2020	South_Dakota	545627.0	53903.0	13400.0	26770.0	13733.0	CrimeData	2025-02-07 03:06:20.701827 U...
26	2020	South_Dakota	545627.0	53903.0	13400.0	26770.0	13733.0	CrimeData	2025-02-07 03:07:32.577832 U...
27	2020	Montana	518813.0	53990.0	13548.0	33529.0	7546.0	CrimeData	2025-02-07 03:07:32.577832 U...

1024	2023	New Mexico	Real gross domestic product	110324.0	GDP	2025-02-07 03:06:11.519505 U...
1025	2024q1	New Mexico	Real gross domestic product	111886.0	GDP	2025-02-07 03:06:11.519505 U...
1026	2024q1	New Mexico	Real gross domestic product	111886.0	GDP	2025-02-07 03:07:25.325293 U...
1027	2022	New Mexico	personal_income_gdp	112181.0	GDP	2025-02-07 03:07:25.325293 U...
1028	2022	New Mexico	personal_income_gdp	112181.0	GDP	2025-02-07 03:06:11.519505 U...
1029	2024q2	New Mexico	Real gross domestic product	112371.0	GDP	2025-02-07 03:07:25.325293 U...
1030	2024q2	New Mexico	Real gross domestic product	112371.0	GDP	2025-02-07 03:06:11.519505 U...
1031	2024q3p	New Mexico	Real gross domestic product	113106.0	GDP	2025-02-07 03:07:25.325293 U...
1032	2024q3p	New Mexico	Real gross domestic product	113106.0	GDP	2025-02-07 03:06:11.519505 U...
1033	2023	New Mexico	personal_income_gdp	117016.0	GDP	2025-02-07 03:07:25.325293 U...
1034	2023	New Mexico	personal_income_gdp	117016.0	GDP	2025-02-07 03:06:11.519505 U...
1035	2023q3	New Mexico	personal_income_gdp	117021.0	GDP	2025-02-07 03:07:25.325293 U...
1036	2023q3	New Mexico	personal_income_gdp	117021.0	GDP	2025-02-07 03:06:11.519505 U...
1037	2023q4	New Mexico	personal_income_gdp	118722.0	GDP	2025-02-07 03:06:11.519505 U...
1038	2023q4	New Mexico	personal_income_gdp	118722.0	GDP	2025-02-07 03:07:25.325293 U...
1039	2024q1r	New Mexico	personal_income_gdp	121270.0	GDP	2025-02-07 03:06:11.519505 U...
1040	2024q1r	New Mexico	personal_income_gdp	121270.0	GDP	2025-02-07 03:07:25.325293 U...
1041	2024q2r	New Mexico	personal_income_gdp	122479.0	GDP	2025-02-07 03:06:11.519505 U...
1042	2024q2r	New Mexico	personal_income_gdp	122479.0	GDP	2025-02-07 03:07:25.325293 U...
1043	2024q3p	New Mexico	personal_income_gdp	123207.0	GDP	2025-02-07 03:07:25.325293 U...
1044	2024q3p	New Mexico	personal_income_gdp	123207.0	GDP	2025-02-07 03:06:11.519505 U...
1045	2022	New Mexico	Current-dollar gross domestic ...	128139.0	GDP	2025-02-07 03:07:25.325293 U...
1046	2022	New Mexico	Current-dollar gross domestic ...	128139.0	GDP	2025-02-07 03:06:11.519505 U...
1047	2023	New Mexico	Current-dollar gross domestic ...	135010.0	GDP	2025-02-07 03:07:25.325293 U...

Transformation

Lower Cap

Underscore for connection

A list of 50 states

Texas – texas

New York – new_york

District of Columbia – REMOVED

Code Snippet

```
transformed AS (  
  SELECT  
    CAST(FLOOR(year) AS INT64) AS year,  
    LOWER(REPLACE(state, ' ', '_')) AS state,  
    population,  
    crime,  
    crime_against_persons,  
    crime_against_property,  
    crime_against_society,  
    _data_source,  
    _load_time  
  FROM source  
  WHERE LOWER(REPLACE(state, ' ', '_')) IN (  
    'alabama', 'alaska', 'arizona', 'arkansas', 'california',  
    'colorado', 'connecticut', 'delaware', 'florida', 'georgia',  
    'hawaii', 'idaho', 'illinois', 'indiana', 'iowa',  
    'kansas', 'kentucky', 'louisiana', 'maine', 'maryland',  
    'massachusetts', 'michigan', 'minnesota', 'mississippi', 'missouri',  
    'montana', 'nebraska', 'nevada', 'new_hampshire', 'new_jersey',  
    'new_mexico', 'new_york', 'north_carolina', 'north_dakota', 'ohio',  
    'oklahoma', 'oregon', 'pennsylvania', 'rhode_island', 'south_carolina',  
    'south_dakota', 'tennessee', 'texas', 'utah', 'vermont',  
    'virginia', 'washington', 'west_virginia', 'wisconsin', 'wyoming'  
  )  
)  
,
```

We don't
need to
use LLM
now

Checking Constraints

FK: Transformed state names

PK: Year & state

Other constraints: not null

```
    enforced: true
constraints:
  - type: primary_key
    columns: [state_name, city_name]
  - type: foreign_key
    columns: [state_name]
    to: ref('Us_Geocode')
    to_columns: [state_full_name]
columns:
  - name: state_name
    data_type: string
    tests:
      - not_null
      - relationships:
          to: ref('Us_Geocode')
          field: state_full_name

  - name: city_name
    data_type: string
    tests:
      - not_null

  - name: avg_density
    data_type: float64

  - name: latitude
    data_type: float64

  - name: longitude
    data_type: float64

  - name: state_id
    data_type: string

  - name: total_population
    data_type: int64
```

Demo Time

The result

 Crime_Data  Query  Open in  Share  Copy  Snapshot  Delete  Export

Schema Details Preview Table Explorer **Preview** Insights Lineage Data Profile Data Quality

 Filter Enter property name or value

☐	Field name	Type	Mode	Key	Collation	Default Value	Policy Tags 	Description
☐	year	INTEGER	NULLABLE	PK	-	-	-	-
☐	state	STRING	NULLABLE	PK	-	-	-	-
☐	population	FLOAT	NULLABLE	-	-	-	-	-
☐	crime	FLOAT	NULLABLE	-	-	-	-	-
☐	crime_against_persons	FLOAT	NULLABLE	-	-	-	-	-
☐	crime_against_property	FLOAT	NULLABLE	-	-	-	-	-
☐	crime_against_society	FLOAT	NULLABLE	-	-	-	-	-
☐	_data_source	STRING	NULLABLE	-	-	-	-	-
☐	_load_time	TIMESTAMP	NULLABLE	-	-	-	-	-

MAY 2024



THANK YOU!

HENG CAI, CHRISTOPHER CHERRY
CS329E - The University of Texas at Austin